

torch.cov#

<https://pytorch.org/docs/stable/generated/torch.cov.html#torch.cov>

Description:

Estimates the covariance matrix of the variables given by the input matrix, where rows are the variables and columns are the observations. A covariance matrix is a square matrix giving the covariance of each pair of variables. The diagonal contains the variance of each variable (covariance of a variable with itself). By definition, if input represents a single variable (Scalar or 1D) then its variance is returned. The sample covariance of the variables xxx and yyy is given by: where $\bar{x}$ and $\bar{y}$ are the simple means of the xxx and yyy respectively, and $\frac{1}{N-1}$ is the correction. If $fweights$ and/or $aweights$ are provided, the weighted covariance is calculated, which is given by:

Function Signature

```
torch.cov(input, *, correction=1, fweights=None,  
aweights=None) → Tensor#
```

Parameters

Keyword Arguments

Returns

Parameters

Keyword

`correction` (int, optional) – difference between the sample size and sample degrees of freedom. Defaults to Bessel’s correction, `correction = 1` which returns the unbiased estimate, even if both `fweights` and `aweights` are specified. `correction = 0` will return the simple average. Defaults to 1. `fweights` (tensor, optional) – A Scalar or 1D tensor of observation vector frequencies representing the number of times each observation should be repeated. Its numel must equal the number of columns of input. Must have integral dtype. Ignored if None. Defaults to None. `aweights` (tensor, optional) – A Scalar or 1D array of observation vector weights. These relative weights are typically large for observations considered “important” and smaller for observations considered less “important”. Its numel must

Returns:

(Tensor) The covariance matrix of the variables.

Code Examples

```
>>> x = torch.tensor([[0, 2], [1, 1], [2, 0]]).T
>>> x
tensor([[0, 1, 2],
        [2, 1, 0]])
>>> torch.cov(x)
tensor([[ 1., -1.],
        [-1.,  1.]])
>>> torch.cov(x, correction=0)
tensor([[ 0.6667, -0.6667],
        [-0.6667,  0.6667]])
>>> fw = torch.randint(1, 10, (3,))
>>> fw
tensor([1, 6, 9])
>>> aw = torch.rand(3)
>>> aw
tensor([0.4282, 0.0255, 0.4144])
>>> torch.cov(x, fweights=fw, aweights=aw)
tensor([[ 0.4169, -0.4169],
        [-0.4169,  0.4169]])
```

Notes & Warnings

NOTE

See also `torch.corrcoef()` normalized covariance matrix.

Detailed Documentation

Documentation

Estimates the covariance matrix of the variables given by the input matrix, where rows are the variables and columns are the observations.

A covariance matrix is a square matrix giving the covariance of each pair of variables. The diagonal contains the variance of each variable (covariance of a variable with itself). By definition, if input represents a single variable (Scalar or 1D) then its variance is returned.

The sample covariance of the variables `xxx` and `yyy` is given by:

where $\bar{x}$ and $\bar{y}$ are the simple means of the `xxx` and `yyy` respectively, and δN is the correction.

If `fweights` and/or `aweights` are provided, the weighted covariance is calculated, which is given by:

where `www` denotes `fweights` or `aweights` (`f` and `a` for brevity) based on whichever is provided, or $w = f \times a$ if both are provided, and

$$\mu_{x*} = \frac{\sum_{i=1}^N w_i x_i}{\sum_{i=1}^N w_i} \quad \mu_x^* = \frac{\sum_{i=1}^N w_i x_i}{\sum_{i=1}^N w_i}$$

$\mu_{x*} = \sum_{i=1}^N w_i \sum_{i=1}^N w_i x_i$ is the weighted mean of the variable. If not provided, f and/or a can be seen as a 1×1 vector of appropriate size.

input (Tensor) – A 2D matrix containing multiple variables and observations, or a Scalar or 1D vector representing a single variable.

correction (int, optional) – difference between the sample size and sample degrees of freedom. Defaults to Bessel's correction, $\text{correction} = 1$ which returns the unbiased estimate, even if both $fweights$ and $aweights$ are specified. $\text{correction} = 0$ will return the simple average. Defaults to 1.

fweights (tensor, optional) – A Scalar or 1D tensor of observation vector frequencies representing the number of times each observation should be repeated. Its numel must equal the number of columns of input. Must have integral dtype. Ignored if None. Defaults to None.

aweights (tensor, optional) – A Scalar or 1D array of observation vector weights. These relative weights are typically large for observations considered “important” and smaller for observations considered less “important”. Its numel must equal the number of columns of input. Must have floating point dtype. Ignored if None. Defaults to None.

(Tensor) The covariance matrix of the variables.

`torch.corrcoef()` normalized covariance matrix.